

1. A user reports their laptop won't power on even when plugged in. List 5 troubleshooting steps you would take to diagnose the cause, and for each step, explain what you are checking for.
 - First, I would check whether the power adapter is working and verify that the power outlet is supplying electricity. Next, I would inspect the charging cable and connector for any physical damage. Then, I would check if the charging LED turns on when the adapter is connected. After that, I would remove the battery (if removable) and try powering on the laptop using only the charger. Finally, I would perform a hard reset by disconnecting the charger and battery, holding the power button for about 30 seconds, and then trying to turn the laptop on again.

2. You receive a helpdesk ticket: 'My monitor is showing a blank screen, but the CPU seems to be running.' Write a step-by-step troubleshooting flow you would follow to identify and resolve the issue.

Hint: Consider checking both hardware connections and basic display settings.

- First, I would make sure the monitor is powered on. Then, I would check that the HDMI, VGA, or DisplayPort cable is connected securely. Next, I would try a different display cable or monitor to rule out a faulty monitor. After that, I would check whether the RAM and graphics card are properly seated inside the computer. Finally, I would restart the PC and verify that the correct display input source is selected on the monitor.

3. A user complains that their wireless mouse is unresponsive, but the keyboard works fine. Describe how you would isolate whether the issue is with the mouse, the USB port, or the system settings.

- First, I would replace the mouse battery or recharge the mouse if needed. Then, I would reconnect the USB receiver and try a different USB port. After that, I would test the mouse on another computer to

see if it works. If it works on another system, I would check the mouse settings and drivers on the original computer. This process helps determine whether the problem is with the mouse, the USB port, or the computer settings.

4. Imagine you are working on a helpdesk team for a gaming cafe. One system is overheating and shutting down during use. List 3 possible hardware causes and describe how you would check each one.

Hint: Think about fans, dust, and thermal paste.

- One possible cause is dust blocking the cooling system. I would open the cabinet and clean the fans and heat sinks using compressed air.
- Another possible cause is a faulty CPU or GPU cooling fan. I would check whether all fans are spinning properly while the computer is running.
- The third possible cause is dried or damaged thermal paste between the CPU and the heat sink. I would remove the heat sink, clean the old thermal paste, apply new thermal paste, and reinstall the cooler properly.